

Procedure to Run Verilog Code in VS Code

1. Prerequisites

Before running Verilog in VS Code, install the following tools:

- Visual Studio Code
- Icarus Verilog (iverilog)
- GTKWave

Install VS Code Extension

- Open VS Code → Extensions
- Install:
 - Verilog HDL/SystemVerilog support extension

3. Project File Structure

Example:

```
verilog-digital-design/  
|  
|—— and_gate.v  
|—— tb_and_gate.v
```

4. Write Verilog Module : example(and_gate.v)

5. Write Testbench : example(tb_and_gate.v)

6. Compile Verilog Code

Open terminal in VS Code and run:

```
iverilog -o output.vvp and_gate.v tb_and_gate.v
```

This compiles both module and testbench

Before compiling and running Verilog files, ensure that the terminal is opened in the correct project directory.

7. Run Simulation

```
vvp output.vvp
```

This generates:

```
wave.vcd
```

8. View Waveform

```
gtkwave wave.vcd
```

You can now see signal transitions graphically

9. Output Explanation

- iverilog → compiles code
- vvp → runs simulation
- .vcd → waveform file
- GTKWave → visualization tool